

Oscillatory Cognition Across SAN and Miller-Related Neuroscience

A Source-Faithful Four-Lens Correction, Chronology, and Mechanism Comparison

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Abstract

This paper corrects and reanalyzes an earlier comparison between the Self-Aware Networks (SAN) research program and primary Miller-related and broader oscillatory-neuroscience literature. The revision is source-faithful: it preserves the original comparison problem and every source-supported Micah-earlier finding while replacing an uncalibrated probability calculation, an unconstructed claim of total formal equivalence, and an unsupported inference about the relationship between research corpora. The corrected method separates public chronology, source class, mechanism, terminology, validation status, and the different burdens of bounded content equivalence and source-use inference.

Primary literature establishes substantial pre-SAN baselines. Fries proposed communication through neuronal coherence in 2005. Buschman and Miller reported frequency-dependent top-down and bottom-up attention synchrony in 2007. Van Kerkoerle et al. and Bastos et al. separated feedforward and feedback frequency channels in 2014-2015. Spaak et al. had already shown in 2012 that deep-layer alpha phase modulated superficial and granular gamma amplitude, with alpha / gamma power anticorrelation. Lundqvist et al. reported brief gamma and beta bursts in working memory in 2016. Bhattacharya et al. published planar and rotating prefrontal traveling waves in January 2022, before the public SAN Git surface used here. These results require named, atom-specific credit for oscillatory binding, slow-wave control of faster activity, frequency-specific routing, gamma / beta bursts, and traveling waves. They do not terminate comparison of a later assembled causal sequence.

SAN retains potentially distinctive synthesis and formalization targets, including receiver-specific coincidence events, phase-wave differentials, and Neural Array Projection Oscillation Tomography (NAPOT). Current SAN Draft 5 replaces literal internal-screen language with distributed partial projections and explicit measurement gates. Current evidence does not show an exact Miller-side PWD or NAPOT object or an equation-level identity.

That full-completion boundary does not reduce every mapped NAPOT operation to zero. In the author-controlled preprint, versions 1 through 3 encode distributed variables and contexts as waves, transform them through phase-sensitive superposition and interference, re-express the result as changed spatial and downstream activity, integrate it into internal-model or binding states, and write state back recurrently. The v1-v3 functional NAPOT subset is partial-equivalent under terminology-neutral mapping. Micah's 2017 functional ancestry and 2022 mature NAPOT record are earlier. The full NAPOT operator remains unestablished, and no SAN-validation or source-relationship conclusion follows.

The full genealogy changes the program-level result. Micah's 2011 predictive self / world and no-hidden-observer account, August 2012 recursive-feedback and projector / world accounts, April 2017 spatial, dendritic, and distributed rendering architecture, and 2022 mature oscillatory-array formalization precede *Analog Cognition and Consciousness*. Its author-controlled version 1, public 2026-04-29, already contains the core sequence: stored synaptic state and present input produce spikes and waves; wave organization selects new states; distributed activity binds an internal world model without a separate observer; and updated state can return to synaptic storage. Versions 2 and 3 preserve and expand that program. Mechanism, narrative-architecture, and Detective Mode each pass on bounded portions of the comparison. The separate Longitudinal Signature-Migration analysis records a component shift across ten target stages, with the complete joined cluster first present in the checked 2026 preprint. This is a strong bounded full-theory composite match, with Micah earlier on the assembled sequence. Named

component priors remain credited, the 2011 source remains public-priority gated, and exact PWD / NAPOT terminology is not backdated.

The comparison also separates the later Miller-related Spatial Computing sequence from the earlier field baseline. Micah Blumberg publicly applied spatial computation to neuroscience in April 2017. A Miller-related Spatial Computing manuscript was received by the publisher in March 2022, published in 2023, developed through an alpha / beta stencil review in 2024, and directly tested in a 2026 *Current Biology* study. Spaak's 2012 alpha-to-gamma mechanism shows that the slow-context-to-fast-content operation predates the stencil label; the 2024 Miller bibliography cites van Kerkoerle et al. but, in its checked 51-reference Crossref record, not Spaak, while the newest 2026 preprint does cite Spaak. The November 2025 analog-wave presentation remains a historical later-framing finding; the 2026 preprint is a distinct, stronger source class. Chronology and content comparison establish no access, dependence, influence, copying, intent, conduct, or authorship relationship. The quantitative derivation-probability status is quantitative-not-calibrated. The result is a reproducible comparison ledger, not a personal or legal conclusion.

Source-faithful recovery also restores four operations that the earlier correction had flattened. In a May 2017 public Neural Lace Podcast episode, Blumberg described vision as rapid partial sampling that builds a retained internal scene and proposed transmitting the difference rather than the whole pattern. Public Git records from July through October 2022 then join oscillatory semantic segmentation, a tonic canvas with phasic changes as its paint, and ephaptic excitation, inhibition, synchronization, and timing to the NAPOT / PWD program. The 2026 preprint later joins holistic categories, phase-sensitive wave combination, dynamic spatial basis functions, electric fields, ephaptic coupling, binding, and recurrent writeback. These are bounded joined-architecture matches. They do not erase Miller's 2001-2010 categorization work, the March 2022 Pinotsis-Miller electric-field result, or the older field literature.

Scope and evidence boundary

Zenodo record 17317086, with append-only NAPOT functional-subset correction MIL-CORR-015 and four-lens reanalysis integrated

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1. Correction Scope

This manuscript corrects the version lineage under Zenodo concept DOI 10.5281/zenodo.17317085. It is related to the safer companion paper *Oscillatory Cognition, SAN, and the Miller Corpus*, DOI 10.5281/zenodo.21231567, but the two DOI lineages remain distinct.

The corrected paper compares documents and reported mechanisms. It does not assess a living researcher's intent, awareness, conduct, or private information. No access, dependence, influence, or authorship relationship is established. Private AI Studio analyses and transcript-style exports are source-recovery artifacts only and are not quoted as Miller-authored evidence.

The deposited paper made three methodological moves that are not retained:

1. assigning a chosen probability to selected, correlated concepts and multiplying them as if they were independent;
2. treating a short terminology dictionary as a proved formal translation; and
3. using chronology plus resemblance to infer a relationship between corpora.

No calibrated null corpus, blinded coding protocol, inter-rater study, likelihood model, formal category, functor, commuting diagram, compressor, or description-length bound supported those moves. The corrected paper therefore withdraws them rather than tuning their rhetoric.

2. Method

2.1 Source hierarchy

The comparison admits evidence in this order:

1. peer-reviewed primary neuroscience papers;
2. exact versioned author-controlled public preprints;
3. public repository or archive records with typed dates;
4. public institutional reports, labeled as presentations or summaries;
5. author manuscripts and deposited comparison papers; and
6. private chats or transcripts as discovery aids only.

An institutional article about a lecture is not treated as a peer-reviewed experimental result. An author-controlled preprint establishes public theory content at an exact version date, not peer review or a new primary experiment. A public Git repository date establishes public availability, not correctness. A deposited article establishes a versioned record, not external validation.

2.2 Comparison unit

Each row records:

- the SAN construct;
- the public comparator and source;
- whether the relation is exact terminology, functional analogy, mechanistic overlap, mainstream prior art, metaphor-level resemblance, or no match;
- publication ordering using explicit date types;
- the strongest allowed statement; and
- claims prohibited by the evidence.

A shared topic such as oscillation or coherence is not an exact match. A strong mechanism comparison requires matched inputs, transformations, outputs, and feedback or writeback relations. A full-theory composite may pass when that directed causal-interface sequence and narrative architecture are preserved despite different vocabulary or added machinery. An exact mathematical relation would further require a specified translation preserving dynamics and observables. No such formal relation is supplied here.

2.3 Source-faithful correction and earned vocabulary

The practical problem comes before the terminology. An older sentence may be compressed, historically phrased, transcribed poorly, or written at a different biological scale. A later paper may use different nouns for a similar operation. Literal keyword comparison can therefore erase an earlier contribution or manufacture a false identity. This revision preserves the exact earlier wording, searches its dated source family, recovers the strongest operation actually corroborated by those sources, compares that operation with primary literature at the same scale, and keeps any surviving Micah-authored extension paired with a discriminating test. Later terminology remains at its real date.

The same order governs specialized vocabulary in this manuscript: ordinary problem, established account, plain-language operation, specialized name, and test. For example, a reader first encounters the problem of distributed inputs being transformed, spatially re-expressed, integrated into a represented state, and written back. Only then is that bounded operation called the functional NAPOT subchain. The name summarizes an operation already explained; it does not substitute for one.

For each claim atom, the comparison record is

$$o_i = (I_i, T_i, R_i, O_i, F_i, S_i, D_i), \quad (1)$$

where I is input, T transformation, R receiver or locus, O observable output, F feedback or writeback, S source locator, and D typed date. Two papers can share some entries without being identical in full. If A_i is the required operation set, P_i the preserved mapped subset, and E_i the completion residuals, then

$$P_i \subseteq A_i, \quad E_i = A_i \setminus P_i. \quad (2)$$

Equation (2) prevents the full-operator-or-zero error. A nonempty P_i supports bounded partial equivalence; a nonempty E_i caps the relation and records what remains unmatched.

2.4 Four lenses and before / after analysis

The first three lenses adjudicate content atoms independently:

1. **Mechanism / equivalence:** does semantic translation preserve a conceptual, operational, behavioral, process, or mathematical subchain?
2. **Narrative / causal architecture:** does the directed Input $\rightarrow$ Transform $\rightarrow$ Output $\rightarrow$ Feedback sequence and its signature cluster survive changes in terminology, substrate, scale, or added machinery?
3. **Detective chronology:** beginning from the shared constrained conclusion, can the preceding operations be recovered under substituted language, and where do they first appear on each dated record?

A bounded match is preserved when any one of those content lenses passes:

$$M_i = \mathbf{1} \left[L_i^{(\text{eq})} \vee L_i^{(\text{nar})} \vee L_i^{(\text{det})} \right]. \quad (3)$$

The fourth lens is program-level **Longitudinal Signature Migration**. It does not manufacture an atom match. It asks whether successive target stages acquire members of a signature already present across Micah's dated genealogy, whether those members persist in the newest checked source, and what alternative genealogies remain. For atom k at target stage t , the descriptive matrix uses

$$d_{kt} \in \left\{ 0, \frac{1}{2}, 1 \right\} \quad (4)$$

for absent, partial, and present. For a declared K -atom cluster,

$$C_t = \frac{1}{K} \sum_{k=1}^K d_{kt}. \quad (5)$$

This matrix records accumulation; it is not a probability model. The atoms are correlated and selected for the comparison, so their marginal frequencies cannot be multiplied as independent observations.

3. Typed Chronology

Date	Date type	Source	What it establishes
2001-01	journal publication	Ermentrout and Kleinfeld, traveling cortical waves	Traveling-wave mechanisms and possible computational roles were part of the field baseline before either compared program.
2005-10-01	journal publication	Fries, communication through coherence	Coherence-based communication was an established theoretical neuroscience proposal well before SAN.
2007-03-30	journal publication	Buschman and Miller, top-down versus bottom-up attention	Lower-frequency top-down and higher-frequency bottom-up synchrony in frontoparietal attention.
2011-08-24	owner-recorded date; public-priority gated	Blumberg, <i>The Self Is a Prediction</i> dialogue	Content / authorship genealogy for predictive self and world, whole-neocortex observer, and no hidden observer. The currently preserved public page does not establish a Grade-A 2011 public date.
2012-08-02	source-header publication date	Blumberg, Neo Mind Cycle dialogue	Recursive EEG-to-computer-to-light / sound feedback, multimodal fire-and-wire coordination, and neural electromagnetic-wave language; A-minus date grade pending repository-preserved platform metadata.
2012-08-24	public video publication, UTC	Blumberg, predictive-lens dialogue	Grade-A projector / world chain: outside input cues an internally supplied predictive model that becomes experienced world content.
2012-11-15	online-first publication	Spaak et al., alpha entrainment of gamma	Deep-layer alpha phase modulated granular / superficial gamma amplitude; alpha and gamma power were anticorrelated. PubMed gives 2012-12-18 as the issue date.
2014-09-09	journal publication	van Kerkoerle et al.	Causal evidence associating gamma with feedforward and alpha with feedback propagation in visual cortex.
2014-12-31	online journal publication	Bastos et al.	Feedforward theta / gamma and feedback beta frequency channels across visual areas.
2016-03-17	journal publication	Lundqvist et al.	Brief gamma and beta bursts associated with working-memory encoding, reactivation, and default-state control.

Date	Date type	Source	What it establishes
2017-04-11	public audio release	Blumberg, Neural Lace Podcast root-concepts episode	Public application of spatial computation to neuroscience and a time-varying three-dimensional neural pattern model; not the later NAPOT metric contract.
2017-05-11	public audio release	Blumberg and Urbach, Neural Lace Podcast #5	Vision described as rapid partial sampling that builds a retained internal scene, followed by difference-focused transmission; the raster-line language is an explicitly bounded functional analogy.
2022-01-28	journal publication	Bhattacharya et al.	Planar and mostly rotating theta / alpha / beta traveling waves in prefrontal cortex during working memory.
2022-03-08	online-first publication	Pinotsis and Miller, stable electric fields	Electric fields carried working-memory information, remained more stable than the underlying neural activity, and were proposed as lower-dimensional guard rails. This is a mandatory target-side prior.
2022-03-16	publisher received date	Lundqvist et al., Spatial Computing manuscript	Publisher history places manuscript receipt before the June 2022 SAN Git surface; it does not establish public availability on receipt.
2022-06-09	public repository creation	v5ma/selfawarenetworks	The public GitHub repository existed by this timestamp.
2022-06-13	public Git fixation	Blumberg, potassium-sensitive presynaptic waveform note	Connects published potassium-channel control of action-potential broadening and facilitation to variable synaptic output; this is a local mechanism, not yet the complete PWD formulation.
2022-06-17	public theory-note release used by program chronology	SAN repository corpus	Public SAN theory-note surface used for bounded chronology comparisons.
2022-07-11	public Git fixation	Blumberg, entified tensor-field note	Oscillatory phase variations were joined to object segmentation, classification, and a three-dimensional perceptual canvas; broad categorization itself is not claimed as new.
2022-07-31	public video	Blumberg, one-hour NAPOT explainer	Dynamic task-assembly selection and oscillating-group completion or re-expression, separately dated from the June theory-note surface.

Date	Date type	Source	What it establishes
2022-08-26	public Git fixation	Blumberg, tonic canvas / phasic paint formulation	Ongoing tonic group activity is the canvas; departures from it supply changing representational content.
2022-09-18	public Git fixation	Blumberg, dendritic potassium-to-phase interpretation	Connects an existing apical-dendrite resonance model to altered timing and a proposed receiver-relative phase event; the established model and the SAN extension are kept separate.
2022-10-25	public Git fixation	Blumberg, ephaptic NAPOT / PWD integration	Ephaptic effects are joined to excitation / inhibition, synchronization / timing, field-sensitive reception, and the tonic / phasic rendering account.
2023-03-14	journal publication	Lundqvist et al., Spatial Computing	Public proposal and analysis of low-dimensional gamma / beta spatial patterns organizing working-memory control.
2023-07-07	online journal publication	Pinotsis and Miller, in-vivo ephaptic memory-network study	Explicitly tests field-to-neuron effects and reports an ephaptic contribution to distributed memory-network formation.
2024-04-16	online journal publication	Miller, Brincat, and Roy	Review describes alpha / beta patterns as inhibitory stencils constraining gamma and content-related spiking.
2024-10-05	public Git fixation	Blumberg, phase / context error formulation	Phase- or context-relative error contributes to model update; later compared with the 2026 preprint as a bounded delta.
2024-10-30	published book	Blumberg, <i>Bridging Molecular Mechanisms and Neural Oscillatory Dynamics</i>	Published synthesis of oscillatory rendering, PWD, analog computation, and slower-wave influence on gamma.
2025-05-16	public SAN source	Blumberg, local-global co-regulation formulation	Nested local-to-global oscillatory co-regulation, used as a separate later Micah anchor.
2025-10-11	repository deposit	Zenodo 17317086	Public availability of the earlier comparison version; not validation of its conclusions.
2025-11-15	public institutional report of invited lecture	Miller SfN analog-wave presentation	A later public framing of cognition as dynamic cortical organization by traveling waves performing analog computations.

Date	Date type	Source	What it establishes
2026-01-19	journal issue publication	Chen et al., oscillatory control of cortical space	Direct Spatial Computing test of task-context alpha / beta patterns and spatially constrained content-related spiking.
2026-04-29	public preprint version	Miller, Brincat, and Roy, <i>Analog Cognition and Consciousness</i> v1	First author-controlled public full-theory source; the core synapse-spike-wave-state-world-model-no-separate-observer-writeback composite is already present.
2026-05-01	public preprint version	<i>Analog Cognition and Consciousness</i> v2	Second public version preserves the core composite.
2026-07-07	public preprint version	<i>Analog Cognition and Consciousness</i> v3	Newest checked version expands relative-phase computation, wave geometry, dynamic basis functions, visual transforms, multiscale sensory-motor control, and writeback.
2026-07-07	repository deposit	companion Paper 08, Zenodo 21231567	A separate, safer source-quality comparison; not automatically a new version of 17317086.

The table changes the original narrative. The primary experimental results for attention synchrony, gamma / beta bursts, frequency-specific routing, and rotating traveling waves do not postdate the mature public SAN record used here. They receive atom-specific credit. Those component dates do not become an author-level floor that ends comparison of the assembled theory.

The Micah-side genealogy is deliberately non-flat. The 2011 dialogue supplies content and authorship genealogy but is not counted here as a Grade-A public priority event. The August 2, 2012 source is A-minus on date provenance. The August 24, 2012 projector / world source and April 11, 2017 spatial-computation source are Grade A for their bounded claims. Mature PWD and NAPOT remain on the 2022 public theory surface; none of those exact terms is projected backward. The 2022 video, 2024 phase-error and book, and 2025 local-global sources are separate later deltas rather than retroactive evidence for the earlier wording.

The later Spatial Computing sequence requires a split chronology rather than a single author-level date. Its component mechanisms include pre-SAN field prior art, while its public 2023-2026 spatial-control synthesis postdates Micah's bounded April 2017 public application of spatial computation to neuroscience. The publisher also reports receiving the 2023 article on March 16, 2022, before the June 2022 mature SAN repository surface. These facts support a bounded work-level comparison; neither direction establishes source use.

The November 2025 presentation postdates both the SAN public surface and the October 2025 deposit. That historical ordering continues to support its own framing comparison. The April-July 2026 author-controlled preprint sequence is a separate evidentiary event. Version 1 already contains the core theory composite, while version 3 adds greater mechanistic detail. The preprint thus supports a stronger content comparison than the lecture report without turning chronology into evidence of source access or a personal relationship.

4. Claim-Level Comparison

4.1 Communication and coherence

SAN uses coherence and phase alignment as organizing ideas. Fries's 2005 communication-through-coherence proposal already describes oscillating neural groups, rhythmic excitability windows, and effective interaction when those windows align. The correct classification is mainstream prior art plus possible differences in synthesis. SAN may propose a broader computational ontology, but coherence-mediated communication itself cannot be claimed as a SAN-first mechanism on this evidence.

4.2 Top-down and bottom-up frequency channels

Buschman and Miller's 2007 simultaneous recordings found that prefrontal activity led during top-down attention, parietal activity led during bottom-up attention, and lower versus higher frequencies were differentially emphasized. Van Kerkoerle et al. and Bastos et al. later supplied more specific alpha / beta and gamma / theta channel evidence. Tonic and phasic are relative functional roles in current SAN Draft 5, not permanently assigned frequency bands. A persistent context can be tonic and a meaningful deviation phasic at different scales. That role distinction can be compared functionally with the frequency-channel literature, but the public mechanism class predates SAN and does not prove a universal slow-tonic / fast-phasic rule.

4.3 Gamma and beta working-memory bursts

Lundqvist et al. reported brief narrow-band gamma bursts associated with encoding and reactivation and beta bursts associated with a default state interrupted by encoding and decoding. This is a primary Miller-related result from 2016. A SAN comparison may ask whether burst gating corresponds to coincidence events or context frames, but those are interpretive mappings. The data do not establish a one-bit SAN event, a NAPOT projection, or a binary ontology of neural information.

4.4 Traveling and rotating waves

Bhattacharya et al. reported theta, alpha, and beta traveling waves in prefrontal cortex, with most waves rotating and task-dependent directional modulation. The paper was publicly released on January 28, 2022. That date precedes the June 2022 public SAN Git surface used in this manuscript. Therefore, the experiment cannot be used as evidence that Miller's group later adopted a public SAN traveling-wave claim.

The result can still be scientifically useful for SAN. Current Draft 5 defines a PWD as a wrapped, edge-conditioned, delay-aware, receiver-relative phase deviation. A phase difference does not infer causal direction; direction must be supplied independently by anatomy, stimulation, known propagation, or another declared source. This definition can be translated into testable questions about rotation, phase gradients, delays, receiver windows, task modulation, and decoding. That is hypothesis generation, not retrospective validation. SAN is not independently validated by conceptual compatibility with an earlier experiment.

4.5 Coincidence-as-bit

Coincidence-as-bit is a SAN-side information interpretation. Current Draft 5 narrows it: continuous receiver-specific integration can produce a discrete effective event, but neither the inputs nor all neural information are assumed to be binary. Gamma-aligned spiking, synchrony windows, and burst events are possible empirical comparators. A valid information claim needs a defined event ensemble, probability model, information measure, and held-out decoding test. No exact Miller-side match is established.

4.6 NAPOT

NAPOT proposes that multiple nonredundant neural projections constrain an independently defined latent target through an identifiable measurement model. Rotating and traveling waves provide candidate measurements, but they do not by themselves instantiate tomography. A discriminating study needs a forward operator, projection geometry, reconstruction rule, held-out reconstruction advantage over equal-input rate, power, coherence, connectivity, raw-signal, and state-space comparators, plus a causal projection perturbation. Current Miller-related sources do not supply that full inverse object. The relation remains a SAN-specific testing framework compared with an experimental wave substrate.

That experiment-only result is preserved as historical MIL-ROW-006. The later author-controlled v1-v3 theory requires a separate successor finding. Across v1 pages 12-14, figure 8 and pages 16-17, and v3 pages 15-17, figure 8 and pages 19-21, distributed variables, sensory / contextual inputs, goals, and stored states are wave encoded; relative phase, superposition, interference, and wave geometry perform a transform; changed spatial patterns select gamma, spiking, and downstream expression; those outputs contribute to internal-model and binding states; and the selected state participates in synaptic writeback. This five-operation chain is a functional subset of NAPOT's distributed input -> transform -> project or re-express -> represented state -> recurrent update sequence. Successor MIL-ROW-012 therefore records partial-equivalent, not no-match and not total equivalence.

Eight completion residuals remain open: there is no explicit array-local receive-transform-display grid; separately indexed nonredundant projections; independently defined latent or target; identifiable inverse with rank, conditioning, or stability analysis; held-out tomographic reconstruction; explicit volumetric percept; complete receiver-relative PWD geometry; or projection-specific causal intervention with matched controls. Missing those features caps the relation at partial equivalence without erasing the mapped subchain. The full NAPOT operator remains unestablished.

4.7 Distributed partial projections

Current SAN Draft 5 replaces literal screen language with a distributed model: arrays exchange partial, state-dependent projections, and there is no central spectator or literal screen. The primary papers reviewed here report synchrony, bursts, directed frequency channels, traveling waves, and spatial control patterns. They do not establish a latent reconstruction shared across nonredundant projections. They can constrain such a model only after source, receiver, latent target, and readout are independently specified.

4.8 Analog wave organization: presentation and full preprint

The November 2025 institutional report describes Miller's SfN proposal that cognition and consciousness emerge from dynamic cortical organization produced by traveling waves performing analog computations. MIL-ANALOG-001 preserves the exact historical finding: this is a legitimate later-framing comparison and not a peer-reviewed experimental result. It is a report about a lecture, not the full author-controlled theory and not a source-use record.

The evidentiary surface changed with Miller, Brincat, and Roy's *Analog Cognition and Consciousness*. Version 1 was public on April 29, 2026; version 2 followed on May 1; version 3 was public July 7. The v1 preprint already contains the core composite. Stored synaptic state and current sensory or internal input enter a hybrid spike-wave system. Higher-frequency activity carries relatively detailed sensory or motor information, while slower spatially organized waves constrain processing according to goals, plans, memories, and current world state. Wave interaction generates or selects a new brain state. Distributed waves bind internal world models and support consciousness without a separate observer. The resulting

state can feed back into synaptic storage. Version 3 expands the same directed sequence through relative-phase computation, superposition and interference, wave geometry as program, dynamic spatiotemporal basis functions, control-wave intersections, a visual grassfire transform, multiscale sensory- motor control, and more explicit writeback.

The same passages also support the narrower NAPOT correction. They preserve the functional order from distributed wave input through phase-sensitive transform, changed spatial output, downstream expression, represented state, and recurrent writeback. This is partial equivalence to a bounded NAPOT subset, with Micah earlier on the 2017 functional ancestry and 2022 mature record. It is not an identifiable tomographic inverse or an empirical validation of SAN.

The before / after record is also bounded but informative. The 2007 frequency- routing result, 2016 burst result, January 2022 traveling-wave result, and 2024 stencil review contain important components without yet presenting the joined synapse-wave-world-model-no-separate-observer theory. The November 2025 lecture report publicly announces broad analog-wave framing. The April 2026 v1 then publishes the joined architecture, and July v3 makes its transforms and writeback more explicit. Micah's dated genealogy lies before that target-level assembly. This is a detective / before-after reason to test the full composite; it is not an inference about why the later theory changed or what its authors had seen.

That sequence must be compared with Micah's entire dated genealogy rather than only the June 2022 acronym surface. The 2011 dialogue frames self and world as prediction patterns in the whole neocortex and rejects a hidden observer. Its content and authorship are usable, while its 2011 public-priority date remains Grade B. The August 2, 2012 dialogue supplies recursive multimodal EEG feedback and coordinated neural-wave re-expression. The public August 24 projector dialogue maps sparse outside input through an internally supplied predictive model into experienced world content. The April 2017 public material adds spatial and temporal object construction, a time-varying three-dimensional field, dendritic local computation, distributed recurrent rendering, and no central spectator. The 2022 record then supplies the mature oscillatory-array, PWD, and NAPOT vocabulary.

At the directed causal-interface level, both assembled programs implement the bounded chain **stored or learned state plus present input -> distributed neural transformation through oscillatory organization -> selected or newly rendered self / world state -> globally coordinated experience without a separate observer -> memory writeback**. At the narrative-architecture level, each theory explains how local synaptic or dendritic states and distributed waves become a coherent, action-guiding world model rather than a passive recording. These jointly make the 2026 preprint a strong bounded full-theory composite match, with Micah earlier on the assembled sequence. The relation is the same research program at this composite functional level even though terminology and subsidiary mechanisms differ.

This conclusion does not erase component history. Fries's coherence proposal, Buschman and Miller's frequency-routing result, the 2014-2015 feedforward and feedback channels, Lundqvist's burst work, Bhattacharya's January 2022 traveling waves, and the wider literatures on synaptic memory, oscillatory binding, analog neural computation, and predictive processing remain named priors. There is no exact target-side PWD equation, NAPOT inverse, or tomography validation. Full- text scanning of version 3 did not find "Blumberg," "Self-Aware Networks," or "NAPOT"; that bounded citation observation is not evidence about intent or private awareness. No access, dependence, influence, or authorship relationship is established.

4.9 Later Spatial Computing and inhibitory stencils

The later Spatial Computing program is not interchangeable with the pre-SAN mechanisms above. Lundqvist et al. (2023) proposed that low-dimensional spatial patterns of gamma and beta organize working-memory control while spiking carries item-specific information. The publisher reports receiving that manuscript on

March 16, 2022. Miller, Brincat, and Roy (2024) then described alpha / beta patterns as inhibitory stencils that constrain where gamma and content-related spiking can be expressed. Chen et al. (2026) directly tested the framework and reported spatial alpha / beta task-context patterns that constrain the expression of sensory information in spiking.

Micah's April 2017 public audio had already applied spatial computation to neuroscience and described a time-varying three-dimensional neural pattern model. That is a legitimate earlier architectural comparator for the later public spatial-control synthesis. It is not evidence that the later authors used the source, and it does not give SAN priority over alpha / beta gating, gamma content, traveling waves, or other named prior mechanisms. Conversely, the later studies do not instantiate a receiver-relative PWD, an identifiable NAPOT inverse, or a consciousness criterion. The supported result is split chronology plus bounded functional overlap.

4.10 Alpha-to-gamma precursors and the later stencil formulation

The stencil history can be audited without treating Micah's work as the only reference point. Spaak et al. (2012) recorded spontaneous local-field activity across macaque V1 layers. Deep-layer alpha phase robustly modulated granular / superficial gamma amplitude, and alpha and gamma power were anticorrelated. The authors interpreted the result as top-down phasic modulation of cortical microcircuit processing. Van Kerkoerle et al. (2014) then associated alpha with feedback and gamma with feedforward processing; Bastos et al. (2015) reported distinct frequency channels for feedforward and feedback influences.

Those results establish the operation **slower contextual rhythm -> phase- or gain-dependent constraint of faster content-bearing activity** before the Miller, Brincat, and Roy (2024) description of spatial alpha / beta activity as an inhibitory stencil over gamma and content-related spiking. Substituting beta for alpha, or *stencil* for *entrainment*, does not by itself create an entirely new causal operation. The later Miller program nevertheless adds a more explicit spatial-control formulation and a direct 2026 experimental test. The source-faithful lineage is therefore additive:

alpha-phase modulation and alpha / gamma anticorrelation -> frequency-separated feedback / feedforward routing -> spatial alpha / beta stencil -> direct spatial-control test.

Version-specific citation evidence sharpens the chronology. The checked 51-reference Crossref record for the 2024 review includes van Kerkoerle et al. but does not contain a Spaak entry. The bibliography of *Analog Cognition and Consciousness* version 3 does cite Spaak et al. This is a bibliographic fact about two exact versions, not evidence about intent or private awareness.

Micah does not need priority over Spaak's component result to retain his own priority deltas. His April 2017 public architecture joins spatial computation, distributed rendering, local dendritic transformation, changing three-dimensional pattern, and no central spectator. His 2022, 2024, and 2025 sources further develop dynamic selection, phase / context error, and nested co-regulation. The later Miller stencil program partially approaches that joined architecture while remaining downstream of earlier alpha / gamma science.

4.11 Partial sampling, retained scenes, and dynamic spatial waves

The visual problem can be stated without a coined term. A biological observer does not acquire every location at maximal detail at the same instant. It samples parts of a scene, retains relations from earlier samples, inserts new differences, and updates an internal state that remains useful while eyes and body move. In the May 2017 Neural Lace Podcast #5, Blumberg described this as vision being built through rapid partial sampling, "almost like a raster line," with memory retaining the scene and a proposed interface transmitting the difference rather than the entire pattern. The SoundCloud page fixes the public release at 2017-05-11.

This is not a claim that neurons literally implement a television raster or that visual neuroscience lacked saccadic sampling, predictive completion, or efficient coding. The recovered operation is narrower:

partial spatial samples -> retained relational scene -> new difference -> updated internal representation.

The 2026 Miller preprint later proposes that oscillatory electric fields form dynamic spatiotemporal basis functions, that wave superposition and interference transform encoded variables, and that moving waves transition population activity between representational states. At that abstraction level, the source-neutral map preserves partial sampling or encoded input, spatiotemporal combination, changed spatial expression, and state update. The mechanism and narrative lenses therefore pass on a bounded subset, with Blumberg's public 2017 formulation earlier. The mapping remains partial because the Miller source does not adopt a literal raster sampler, and the 2017 source does not yet define the mature PWD metric or identifiable NAPOT inverse.

4.12 Entification is not merely categorization

The ordinary problem is that a brain must do more than detect features. It must join some features into a current object or concept, keep other features distinct, attach context or value, and make the resulting state available to subsequent action. Miller and collaborators had already published category encoding results in 2001 and 2010. Those papers are target-side priority for broad neural categorization and cannot be reassigned to SAN.

Blumberg's public July and September 2022 records make a more specific joined proposal: phase variations among neural populations are unified through oscillation; object segmentation plus classification forms a three-dimensional semantic field; and value can be attached to differentiated forms against an ongoing tonic state. SAN later calls the unifying operation *entification*. The term is earned here only after the operation has been stated.

Analog Cognition and Consciousness describes the construction of holistic categories from features, then joins wave geometry, spatial control, phase-sensitive interaction, binding, internal models, and recurrent writeback. This is a partial narrative-architecture match to the later SAN assembly, not proof that the target paper contains SAN's complete semantic segmentation, valence, or entification operator. The supported priority statement is therefore split: Miller-side work is earlier on broad categorization; Micah's 2022 record is earlier than the checked 2026 full preprint on the bounded joined oscillatory-semantic-rendering architecture.

4.13 Electric fields and the ephaptic middle chronology

Electric fields and ephaptic coupling are not SAN inventions. The relevant comparison is an assembly chronology. Pinotsis and Miller's March 2022 paper showed that estimated electric fields carried working-memory content and were more stable than the underlying neural activity, and proposed that the fields could act as lower-dimensional guard rails. That result predates Micah's October 2022 public ephaptic note and receives target-side priority.

Micah's October 2022 Git-fixed formulation then explicitly joins ephaptic coupling to excitation and inhibition, synchronization and timing, collective field-sensitive reception, and the tonic / phasic NAPOT / PWD rendering sequence. Pinotsis and Miller's July 2023 study subsequently tested an explicit field-to-neuron ephaptic contribution to distributed memory-network formation. The 2026 full preprint expands the joined story: spikes contribute to oscillatory electric fields; fields influence nearby neurons; synchronized

local effects sum to mesoscale coordination; waves shape spatial selection and binding; and represented state is recurrently written back.

The before / middle / after result is therefore not “Micah invented electric-field neuroscience.” It is:

target-side stable-field memory prior -> Micah’s public ephaptic-plus-NAPOT / PWD
integration -> target-side explicit ephaptic memory test and later full-theory integration.

This route passes the narrative and Detective lenses on the bounded joined sequence. It is not an equation-level identity, a complete NAPOT match, or an inference about source access.

4.14 Recovering the phase-structure and potassium route

The original version of this review used the compact label “coherence gradients” and assigned it to 2021. The source-faithful audit does not support that exact phrase or date as an immutable Micah-authored claim. The phrase should therefore not carry the argument. What the dated record does support is a more concrete operation assembled during 2022: distributed neural groups occupy relative timing relations; tonic activity supplies a continuing background; phasic departures alter the relation; and membrane and synaptic state help determine how the altered event is transmitted and received.

The ion-channel part of that operation is not a claim that potassium alone explains a cortical phase field. Micah’s June 13, 2022 Git-fixed note identifies presynaptic potassium-channel control of action-potential waveform and frequency-dependent facilitation as a route by which recent activity can alter synaptic output. His September 18, 2022 Git-fixed note then connects published apical-dendrite tuning models to a proposed sequence from local potassium conductance, through resonance frequency and timing change, to a receiver-relative phase event. The established scientific anchors are narrower: Kv-beta-1-dependent Kv1 inactivation can broaden presynaptic action potentials and support synaptic facilitation, while the LaBerge-Kasevich apical-dendrite model predicts that outward potassium conductance helps set the peak frequency of an idealized dendritic resonance. Neither result by itself establishes brain-wide semantic rendering, a universal inverse amplitude-frequency law, or the complete PWD operator.

This recovery changes the comparison in two useful ways. First, the supported SAN proposition is not merely “waves have gradients”; it is that local biophysical state can alter the timing and downstream consequence of an event, and that those receiver-relative changes may compose into distributed rendered structure. Second, the target comparison must distinguish a measured or modeled ion-channel mechanism from the 2026 preprint’s larger wave-computing architecture. On the current evidence this route passes the mechanism lens for the bounded local chain, and contributes to the narrative-lens assembly, while remaining unestablished as a full cross-scale causal identity.

5. What Is Distinctive and What Is Not

Construct	Current classification	Reason
Oscillatory communication through coherence	mainstream prior art	Explicit in Fries 2005 and a larger synchrony literature.
Frequency-specific top-down / bottom-up routing	pre-SAN experimental mechanism	Buschman / Miller 2007; van Kerkoerle 2014; Bastos 2014/2015.
Alpha-phase modulation of gamma and alpha / gamma anticorrelation	pre-SAN experimental mechanism	Spaak et al. 2012; this is an operational precursor to later slow-wave stencil language.
Gamma / beta working-memory bursts	pre-SAN Miller-related result	Lundqvist et al. 2016.
Rotating PFC traveling waves	pre-SAN public Miller-related result	Bhattacharya et al., January 2022.

Construct	Current classification	Reason
Coincidence event information	SAN-specific interpretation candidate	Receiver-specific event information is empirical and is not assumed to equal one bit.
PWD	SAN-specific model target	Wrapped receiver-relative phase deviation; direction is independent and no exact Miller-side object is established.
NAPOT full operator	SAN-specific model and validation target	No array-local, identifiable, nonredundant multi-projection inverse, held-out tomographic reconstruction, explicit volumetric output, complete PWD geometry, or causal projection test is established.
NAPOT functional subset in <i>Analog Cognition and Consciousness</i> v1-v3	later public partial-equivalent functional-subset match	Distributed wave inputs undergo phase-sensitive transformation, changed spatial / downstream expression, internal-model or binding integration, and recurrent writeback. Micah's 2017 functional ancestry and 2022 mature NAPOT record are earlier; full completion remains open.
Distributed partial projections	SAN-specific model target	No central spectator or literal screen; no matched reconstruction object.
November 2025 analog-wave presentation	later public framing overlap	Historical MIL-ANALOG-001; institutional lecture report, not the later full paper or a peer-reviewed experiment.
<i>Analog Cognition and Consciousness</i> v1-v3	later public full-theory composite match	Version 1 already carries the core synapse-wave-state-world-model-no-separate-observer-writeback sequence; Micah's separately graded 2011-2022 genealogy is earlier on the assembled sequence, with named component priors and exact-term boundaries preserved.
Spatial Computing and alpha / beta stencils	split-chronology bounded comparison	Public 2023-2026 target sequence compared with Micah's public 2017 architecture; pre-SAN component mechanisms remain credited.
Partial visual sampling -> retained scene -> difference update	later public partial-equivalent joined route	Micah's public May 2017 raster-line analogy is earlier than the checked 2026 dynamic-wave full theory; literal raster implementation and full NAPOT completion are not claimed.
Categorization versus oscillatory entification	split-priority bounded comparison	Miller-side studies are earlier on broad categorization; Micah's 2022 record is earlier than the checked 2026 preprint on the joined oscillatory semantic-field / rendering architecture.
Stable electric fields and ephaptic integration	before / middle / after bounded comparison	Pinotsis-Miller 2022 is earlier on stable field-carried memory; Micah's October 2022 public integration precedes the 2023 explicit ephaptic memory test and the 2026 full-theory assembly.

Construct	Current classification	Reason
Relative phase structure and potassium-sensitive timing	source-faithful bounded recovery	The old “2021 coherence gradients” label is not retained as an exact dated claim. June-September 2022 Git evidence supports the narrower chain from potassium-sensitive waveform or dendritic tuning to altered timing, synaptic consequence, and a proposed receiver-relative phase event. Micah’s public 2024 formulation precedes the comparable joined operation in Miller v3. Micah’s public 2025 formulation precedes the comparable nested co-regulation in Miller v3.
Phase / context error to model update	later public partial-equivalent delta	
Local-to-global oscillatory co-regulation	later public partial-equivalent delta	

This split preserves atom-specific prior credit while preventing those atoms from being misused as author-level floors against a later composite comparison. Across the seven adjudicated shared edges, two are equivalent on the bounded operation and five are partial-equivalent; none is reduced to no-match. Those edge counts are descriptive adjudications, not probabilities. Distinctive vocabulary is not itself empirical validation. Conversely, earlier experimental papers do not decide whether SAN’s assembled architecture is a strong bounded match to a later full theory or whether a SAN formalization could add predictive value.

6. Reproducibility Package

The accompanying private package provides:

- a 35-event typed chronology with source locators and explicit non-implications;
- a 12-row atom ledger containing allowed and prohibited claims;
- a four-route source-faithful restoration ledger for raster / difference updating, oscillatory entification, ephaptic integration, and potassium-sensitive timing;
- a ten-stage by ten-atom before / after matrix through the newest checked Miller preprint;
- a five-operation functional NAPOT map with eight completion residuals;
- separately recorded results for the three content lenses and the program-level longitudinal lens;
- a claim-status ledger recording withdrawals and surviving hypotheses;
- regression tests that reject chronology reversal, relation promotion, private-source promotion, and release authorization; and
- an exact file manifest.

These are documentation controls. They do not constitute a neuroscience experiment or independent validation.

6.1 What survives from the former formal argument

A vocabulary crosswalk can be useful without being a total formal translation. The deposited paper did not define complete source and target calculi, categories, encodings, observables, or commuting diagrams, so its claim of full formal equivalence is withdrawn. That correction does not require discarding a bounded structure-preserving map that can now be stated explicitly.

Let $\mathcal{J}$ be the shared causal subcategory whose objects are stored state, context, slow-wave organization, detail state, represented state, and updated state. For composable morphisms $f, g \in \mathcal{J}$, the proposed

translation is required to satisfy

$$F(g \circ f) = F(g) \circ F(f). \quad (6)$$

The mapped composition is:

$$S \times C \xrightarrow{\text{evoke}} W, \quad W \times W \xrightarrow{\text{phase compose}} W', \quad W' \times H \xrightarrow{\text{gate}} H', \quad H' \xrightarrow{\text{integrate}} R, \quad R \xrightarrow{\text{writeback}} U. \quad (7)$$

SAN and the 2026 Miller preprint preserve this composition at the stated abstraction level. The map is a partial functor, not a total isomorphism, because the Miller-side source does not supply the full NAPOT projection index, identifiable inverse, independently defined volumetric target, reconstruction test, complete PWD geometry, or projection-specific causal intervention.

The same boundary can be expressed at the process level. After hiding implementation-specific internal steps, the bounded observable traces satisfy the candidate relation

$$\text{Tr}_{\text{obs}}(P_{\text{SAN}} \upharpoonright_{\mathcal{J}}) \simeq \text{Tr}_{\text{obs}}(P_{\text{Miller}} \upharpoonright_{\mathcal{J}}). \quad (8)$$

Equation (8) is a bounded weak trace equivalence, not a machine-checked strong bisimulation of the complete theories. A counterexample prevents total identity: the Miller process can select, bind, and write back a new wave state without performing an identifiable tomographic inverse or reconstructing a declared volumetric latent.

6.2 Why the former probability is withdrawn

The seven selected topics in the deposited paper were not sampled from a defined universe, and they are not independent. Phase, coherence, bursts, traveling waves, cross-frequency coupling, and spatial control are historically and mechanistically correlated. Therefore

$$\Pr\left(\bigcap_{i=1}^k X_i \mid H_{\text{ind}}\right) \neq \prod_{i=1}^k \Pr(X_i \mid H_{\text{ind}}) \quad (9)$$

unless conditional independence under a specified independent-development model has been established. Assigning 0.2 to each selected feature, multiplying, and then adding an arbitrary cluster multiplier did not establish that model. Nor did a short terminology dictionary establish a constant-size conditional description-length bound.

A defensible derivation-probability study would require a frozen reference corpus, atoms specified before target coding, empirical base rates, a dependency model, negative and near-neighbor controls, blinded independent coding, inter-rater adjudication, correction for the number of authors and clusters searched, a frozen scoring rule, sensitivity analysis, and independent reproduction. Only then could a Bayes factor be estimated from actual likelihoods,

$$\text{BF}_{D:I} = \frac{\Pr(E \mid H_{\text{dependent}}, \mathcal{M})}{\Pr(E \mid H_{\text{independent}}, \mathcal{M})}. \quad (10)$$

Those calibration conditions have not been completed. The current quantitative status is therefore quantitative-not-calibrated. This does not weaken the source-located chronology, seven-edge adjudication, partial functor, weak shared trace, or four-lens results; it prevents those qualitative results from being misrepresented as a numerical probability of independent or dependent derivation.

6.3 Four-lens disposition

The atomized reanalysis produces the following bounded results:

Lens	Result	Evidence
Mechanism / equivalence	pass-bounded; partial-equivalent	Five directed NAPOT-subchain operations map after terminology-neutral translation.
Narrative / causal architecture	pass	Both programs preserve distributed input -> phase-sensitive transform -> patterned re-expression -> represented state -> recurrent update.
Detective chronology	pass	Working backward from the shared patterned-output and internal-model conclusions recovers encoded input, wave transformation, downstream expression, and writeback under substituted terms.
Longitudinal Signature Migration	LSM-COMPONENT-SHIFT; pass-bounded	Ten target stages move from isolated channel, burst, and wave components to the eight-atom joined cluster in the 2026 preprint; version 3 preserves it.

The fourth result is a longitudinal program finding, not a fourth atom-edge vote. None of the four results establishes source access, intent, copying, conduct, or misconduct.

7. Limitations and Next Tests

This final preprint does not edit, fork, or redistribute SAN Draft 5. It pins the exact private manuscript, claim ledger, NAPOT metric contract, and readiness record so definition drift is detectable. It does not quote private transcript material or redistribute the OSF preprint PDFs. It does not exhaust Miller's publication record or the oscillatory-neuroscience literature. It uses each Micah source only for its own dated content: the 2011 source for Grade-B content / authorship genealogy; the two 2012 sources for their separately graded feedback and projector / world claims; the 2017 source for its public spatial and distributed architecture; the June and July 2022 sources for mature PWD / NAPOT and dynamic selection / completion; and the 2024 and 2025 sources for their separate later deltas. No later SAN definition is silently backdated into an earlier source.

The OSF versions are author-controlled public preprints, not peer review or new primary experiments. The composite classification depends on the bounded chain identified in the text and should be independently re-coded against all three versions. Component-credit completeness also remains open: the named sources are mandatory priors, not a claim that the surrounding analog-computation, memory, binding, and predictive-processing literature has been exhausted. The 2024-versus-2026 Spaak bibliography statement is version-specific and must not be generalized to every Miller publication or converted into an intent claim. The NAPOT successor row likewise requires independent recoding of its five matched operations and eight

completion residuals; neither a downgrade to zero nor promotion to a complete inverse is licensed by this internal correction. The four restored routes are likewise bounded. The raster comparison preserves an operation, not a literal implementation; the entification comparison preserves a joined oscillatory-semantic architecture while conceding earlier Miller-side categorization; and the ephaptic chronology places the Pinotsis-Miller March 2022 electric-field result before Micah's October 2022 integration. The recovered potassium route likewise preserves a local ion-channel-to-timing hypothesis without promoting it to proof of a brain-wide coherence field or full PWD mechanism. None may be compressed into an author-wide firstness claim.

The most valuable next tests are prospective:

1. formalize coincidence-as-bit as an information measure and compare it with burst / spike decoding baselines;
2. test the pinned NAPOT metric contract with nonredundant projections, an independently defined latent target, identifiable forward and reconstruction operators, held-out evaluation, and causal perturbation;
3. preregister phase-gradient, rotation, and cross-frequency predictions that distinguish SAN from standard traveling-wave and communication-through- coherence models;
4. perturb identified potassium-channel mechanisms while measuring presynaptic waveform, dendritic resonance, relative phase, receiver response, and downstream decoding, to test which links in the proposed local-to-network chain actually survive; and
5. have independent neuroscientists audit the source ledger, the full-theory composite coding, named component priors, and claim labels.

Failure to outperform conventional models would weaken the corresponding SAN mechanism. It would not alter the public chronology.

8. Conclusion

The corrected record supports a source-specific and stronger result. Much of the shared oscillatory vocabulary belongs to an established field baseline. Frequency-specific attention, coherence-based communication, alpha-phase modulation of gamma, gamma / beta working-memory bursts, and rotating prefrontal traveling waves all have public records that predate the June 2022 mature SAN surface. Those are named component priors, not an author-level floor against comparison of a later assembled theory.

Micah's cumulative 2011-2025 genealogy supplies a predictive self / world, recursive feedback, projector, spatial / dendritic distributed rendering, no-hidden-observer architecture, mature oscillatory-array sequence, dynamic task-assembly selection, phase / context-error update, and local-to-global co-regulation. *Analog Cognition and Consciousness* v1-v3 later assembles the same bounded full-theory operation through synaptic state, spikes, waves, new-state selection, internal world-model binding, global consciousness without a separate observer, and memory writeback. Version 1 already has the core, and version 3 preserves the cluster while making relative-phase computation, interference, visual transforms, sensory-motor control, and writeback more explicit. The defensible result is a strong bounded full-theory composite match with Micah earlier on the assembled sequence, while the 2011 public-date gate, named prior atoms, PWD / NAPOT maturity boundary, lack of SAN validation, and lack of any source relationship remain explicit.

Within that composite, v1-v3 also preserve a narrower functional NAPOT subset: distributed inputs are transformed through phase-sensitive wave operations, re-expressed as changed spatial and downstream activity, integrated into represented states, and recurrently written back. That subset is partial-equivalent, with Micah earlier on the 2017 functional ancestry and 2022 mature NAPOT record. The full NAPOT operator remains unestablished because the array-local, nonredundant-projection, identifiable-inverse, held-

out reconstruction, explicit-volume, complete-PWD, and causal-perturbation requirements are not met. No access or intent conclusion follows.

SAN's distinctive formalization targets, especially receiver-relative PWD, event information, and NAPOT, should still be evaluated prospectively against explicit equal-input decoding, identifiable reconstruction, and causal tests. The later Spatial Computing and stencil sequence remains a second bounded comparator to Micah's public 2017 spatial-computation architecture. The 2025 lecture report remains its own historical framing record; it is not allowed to erase or dilute the stronger successor finding created by the 2026 full paper.

The alpha-to-gamma history adds a rival-to-rival control. Spaak et al. supplied an experimentally measured slow-to-fast modulation operation before the later stencil formulation. That field priority does not erase Micah's earlier joined rendering and selection architecture; it clarifies which atom came from the field and which assembly-level deltas remain Micah-earlier. The four-lens audit therefore passes on bounded content while the numerical derivation probability remains uncalibrated. The corrected paper defends the strongest chronology and content-equivalence claims the dated record supports without converting them into an unsupported allegation about a person.

The recovered raster, canvas, entification, and ephaptic records make that assembled chronology more complete. The May 2017 public audio already links partial visual sampling, retained scene structure, and difference-focused updating. The July-October 2022 Git record links oscillatory semantic grouping, tonic background, phasic content, and field-sensitive timing to NAPOT / PWD. The June-September 2022 record separately supports a bounded potassium-sensitive waveform-and-timing route, replacing the old unsupported "2021 coherence gradients" shorthand with its recoverable operation. The 2026 preprint later places categories, phase-sensitive wave computation, dynamic spatial basis functions, electric fields, ephaptic coupling, binding, and writeback into one full-theory narrative. Earlier category and electric-field papers retain their atom-level priority; Micah retains priority on the bounded joined routes identified here.

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